

4. On the Invariant Theory of Forms in n Variables^[118 : *].

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Editorial note: The leading “4.” is the paper number in this project collection; it is not part of the title printed in the 1911 journal. This checkpoint contains the Introduction only.

Introduction.

In the projective invariant theory of forms in n variables, the questions associated with the main problem — proving the finiteness of the system of forms — have scarcely been treated once one passes beyond the binary and ternary domains. These are the questions of the interdependence of the forms and of the methods for handling these interrelations. As has been fully established in the binary and ternary domains, these methods rest essentially on two theorems that Study calls the “fundamental theorems of the symbolic method”: the theorem on the symbolic representability of forms and the theorem on symbolic identities.^[118 : **] The latter theorem states that all relations among integral invariant constructions are obtained by successive application of a small number of symbolic identities; thus the symbolic method, and it alone, gives knowledge of the interrelations among the forms without leaving the invariant domain.

Gordan had already identified the reason for the difficulty of passing to n variables in his Erlangen program:^[119 : *] the first theorem of the symbolic method no longer holds in its general form. As is well known, higher-grade variable rows p_ρ ^[119 : **] occur in the n -ary domain, and analogously higher-grade symbol rows S^ρ . One arrives at these rows both when, as Clebsch originally did,^[119 : ***] one starts from forms with arbitrarily many rows p_ρ , and when, following F. Deruyts and A. Capelli,^[119 : †] one starts from forms containing only arbitrarily many cogredient rows x . No symbolic representation exists that would contain the rows p_ρ and S^ρ explicitly;^[119 : ††] it is necessary to resolve p_ρ and S^ρ into the individual rows x and the rows s contragredient to x , and to distribute these among different determinants. Differential conditions (cf. formula (19)) do make it possible to verify that a given aggregate of determinants depends only on the rows p_ρ and S^ρ ; but this gives no overview of the totality of invariant constructions or of the processes that generate them.^[119 : †††]

We now show in §§ 2 and 6 how, by applying the “theorem on corresponding matrices,”^[119 : *†] the first fundamental theorem is recovered in full generality by replacing representation by determinants with representation by “matrix products.” Section 2 shows that every matrix product explicitly containing the rows S^ρ and p_ρ is an invariant construction of the ground form. The converse proved in § 6 — that every invariant construction can be represented explicitly by matrix products — is obtained only by transferring the second fundamental theorem (§§ 3–5).

^[118 : *]Cf. a short communication to the Salzburg Meeting of Natural Scientists, *Jahresbericht der Deutschen Mathematiker-Vereinigung*, vol. 19, p. 101.

^[118 : **]E. Study, *Methoden zur Theorie der ternären Formen* (Leipzig: Teubner, 1889), II, § 6. For the occurrence of these fundamental theorems also in the invariant theory of special transformation groups, cf. Study, *Das Apollonische Problem*, *Math. Ann.* 49 (1897), and *Zur Differentialgeometrie der analytischen Kurven*, *Transactions of the American Mathematical Society* 10 (1909), pp. 1–49. The original prints vol. 1.

^[119 : *]P. Gordan, *Über das Formensystem binärer Formen*, p. 46 (Leipzig: Teubner, 1875).

^[119 : **]For the notation, see § 1.

^[119 : ***]A. Clebsch, *Über eine Fundamentalaufgabe der Invariantentheorie*, *Abhandlungen der Gesellschaft der Wissenschaften zu Göttingen* 17 (1872).

^[119 : †]Cf. A. Capelli, *Lezioni sulla teoria delle forme algebriche* (Naples, 1902), §§ 22–24, and the literature cited there.

^[119 : ††]A more precise definition of “explicit representation” is given in § 2.

^[119 : †††]A computationally very valuable extension of the symbolism due to Study — communicated by F. Engel, *Berichte der Königlich Sächsischen Gesellschaft der Wissenschaften zu Leipzig*, mathematical-physical class (1900), p. 126, and developed for the quaternary domain by E. Study, *Geometrie der Dynamen*, p. 135 — is likewise not an explicit representation in our sense. It would, for example, hardly lead to the nonsymbolic differentiation processes that generate the forms.

^[119 : *†]Apart from Grassmann’s *Ausdehnungslehre* of 1862, no. 112, the theorem first appears in A. Brill, *Über zwei Berührungsprobleme*, § 2, *Math. Ann.* 4 (1871).

The latter theorem is known for forms containing only variable rows x .^[120 : *] The general problem of setting out all indecomposable identities in which every row S^ρ and p_ρ is treated as indecomposable was formulated by Study;^[120 : **] he also regarded the number of identities that occur as a measure of the difficulty of the methods in the n -ary domain. By combining all these identities in a single formula, (36), we reduce this difficulty at least to a minimum. In applications, however, certain distinguished special cases of (36) will frequently occur, such as (20) and (21), which are characterized by admitting explicit representation without the substitution of new rows, and formula (31), called the “decomposition identity” because a row p_ρ appears to be decomposed into rows y .

The further theory can now readily be built on the two fundamental theorems. The first theorem directly supplies the processes for generating forms: the symbolic folding process and the nonsymbolic differentiation processes,^[120 : ***] while the decomposition identity eliminates all equivalent processes among them (§ 6). Knowledge of the differentiation processes further extends the concept of the “normal form” to the n -ary domain and, by means of a proof method given by Mertens in the quaternary domain,^[121 : *] yields the theorem that a system of invariant constructions can be replaced by an equivalent system of normal forms (§ 7). Under this latter assumption I showed in my dissertation^[121 : **] that the ternary folding process can be generated by successive application of two fundamental foldings. On the basis of this theorem and the theory of series expansions, I then introduced the concept of a “form series” and developed the principal reduction theorems for systems of forms. Entirely analogously, in the n -ary domain the folding process can be generated by $n - 1$ fundamental foldings (§ 8), and the reduction theorems can be established (§ 9).

Postscript. On the occasion of the Salzburg communication, Professor E. Müller of Vienna drew my attention to the fact that the calculation with matrix products underlying this paper is, in principle, identical with the calculus of Grassmann’s theory of extension.^[121 : ***] Indeed, Grassmann’s “combinatorial product” $[S^{\rho_1} S^{\rho_2} \dots S^{\rho_i}]$ can be regarded as a matrix determined by these rows (cf. § 2): the “progressive product” as the matrix of the rows themselves, the “regressive product” as the matrix of the associated rows $S_{n-\rho_1}, S_{n-\rho_2}, \dots, S_{n-\rho_i}$, and the matrix corresponding to the “mixed product” as the one obtained when several rows S are combined by substitution (9) into new rows P and Q . Our “associated row” corresponds to Grassmann’s “complement,” and our “matrix product” to a “mixed product of grade 0.”

Under this correspondence, the expansion given in the *Ausdehnungslehre* of 1862, no. 113, becomes identical with a formula dual to our formula (32). In the special case $\rho = 1$, formula (32) becomes (17), while the dual formula becomes (16). Müller^[122 : *] recently derived the identities (20) and (21) mentioned above from these special cases of Grassmann’s expansion.

Unlike the *Grassmann–Müller* derivation, ours simultaneously proves the *completeness* of the identities, which appears to be the essential point for the questions considered here; and it advances from (20) and (21) to the most general indecomposable identities (36), which do not appear to have been observed before.

^[120 : *]E. Pascal, *Giornale di matematiche* 26 (1888), pp. 33, 102; *Rendiconti della Reale Accademia dei Lincei* (4) 4 (1888), p. 119; *ibid.*, *Memorie* (4) 5 (1888), p. 375.

^[120 : **]Study, *Methoden*, p. 204, note 17.

^[120 : ***]In the paper *Invariante Gebilde von Nullsystemen*, *Sitzungsberichte der Akademie der Wissenschaften in Wien* 97, part IIa (1888), Mertens obtained the quaternary folding process by another route that does not generalize directly. The alternating invariant of six rows S^2 that occurs there differs from the one obtained here by a single application of substitution (11) by an aggregate of even invariants. A special case of the quaternary folding process is also found in P. Gordan, *Das simultane System von zwei quadratischen quaternären Formen*, *Math. Ann.* 56 (1903).

^[121 : *]F. Mertens, *Über invariante Gebilde quaternärer Formen*, *Sitzungsberichte der Akademie der Wissenschaften in Wien* 98, part IIa (1889).

^[121 : **]*Über die Bildung des Formensystems der ternären biquadratischen Form*, this journal, vol. 134 (1908).

^[121 : ***]*Hermann Grassmanns gesammelte mathematische und physikalische Werke*, vol. 1, part 2: *Die Ausdehnungslehre von 1862*, edited by Fr. Engel in collaboration with H. Grassmann the Younger (Leipzig: Teubner, 1896).

^[122 : *]E. Müller, *Beiträge zur Grassmannschen Ausdehnungslehre*. First communication: *Einige allgemeine Sätze*, *Sitzungsberichte der Akademie der Wissenschaften in Wien* 118, part IIa (July 1909), theorems 16 and 18.